

Thiamine (Vitamin B1) Deficiency

What is it?

Thiamine (vitamin B1) helps the body turn food into energy. The brain is especially sensitive to low levels. Without enough thiamine, brain cells cannot make enough energy, causing neurological signs that can **worsen quickly**.

Causes

Usually related to diet:

- **Unbalanced diets** — home-made or diets not labeled "complete and balanced"
- **All-fish diets in cats** — some raw fish contain an enzyme (thiaminase) that destroys thiamine
- **Overcooked food** — heat destroys thiamine
- Preserved meats containing **sulfur dioxide** (sulfites)
- Rarely, certain drugs or toxins; also poor absorption from intestinal disease

Signs

- Balance problems — wobbliness, stumbling, falling
- **Head tilt** or abnormal eye movements
- Head and neck bent downward (especially cats)
- Dullness, sleepiness → can progress to **coma**
- Dilated pupils or vision changes
- Tremors, **seizures**

Is it an emergency?

It can be. Contact your veterinarian **right away** for sudden balance problems, abnormal eye movements,

severe lethargy, seizures, or rapid worsening.

Diagnosis

- Neurological exam and a **detailed diet history** (food, treats, supplements, home-made components)
- Blood work to rule out other causes
- **MRI** in some cases
- **Rapid improvement after thiamine treatment** strongly supports the diagnosis

Treatment

- **Thiamine supplementation** — injections at first in serious cases, then by mouth
- **Switch to a complete and balanced diet**
- Supportive care: fluids, anti-nausea and anti-seizure medication, hospitalization if needed

Prognosis

Good with prompt treatment. Many pets improve within days and return close to normal within about 2 weeks. Delayed treatment can lead to permanent damage.

Prevention

- Feed a commercial **complete and balanced** diet for your pet's life stage
- Home-cooked diets should be formulated by a **veterinary nutritionist** (BalanceIT.com, petdiets.com)
- Avoid long-term all-fish diets, especially in cats